Causal Co-Scaling of Distance and Time in a Fractal S^3 Reconstruction Model

A Research Note on the Invariant Ratio $R/t = c$

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Manuscript scope. This research note isolates one structural axiom used in the broader compact S^3 programme: distance, reconstruction time, and the speed of light are not independent quantities, but co-scaled components of one causal package.

Abstract. This note formulates an axiom of *causal co-scaling* for fractal compact S^3 reconstruction. The speed of light is not treated as an isolated number to be derived after distance and time have already been defined. Instead, the causal ratio

$$c = \frac{R}{t}$$

is taken as the invariant relation binding distance and reconstruction time. If a Planck-normalised hierarchy scales carrier radii as $R_N = \ell_P \varphi^N$, then causal self-consistency requires reconstruction times to scale as $t_N = t_P \varphi^N$. The ratio is then preserved at every level:

$$\frac{R_N}{t_N} = \frac{\ell_P}{t_P} = c.$$

This does not derive the numerical value of c, since both ℓ_P and t_P already encode the empirical causal scale. Rather, it states a closure condition: a golden-ratio reconstruction hierarchy is causally admissible only when distance and time co-scale. The note also separates the pre-photon reconstruction regime from the later photon-supported cosmological expansion regime, and clarifies the status of an internal transition index such as $N_* = 207$. Such an index is not an observational definition of recombination, but a model-internal reference scale at which the reconstruction mechanism is allowed to change regime.

Keywords: causal co-scaling; compact S^3; golden-ratio hierarchy; Planck scale; reconstruction time; speed of light; fixed-point closure.

1. Purpose and Claim Status

The purpose of this note is narrow. We isolate a single structural axiom:

> *If a carrier-distance scale is multiplied by a reconstruction factor, the corresponding reconstruction time must be multiplied by the same factor, or else the causal ratio R/t is not preserved.*

The note therefore does **not** claim:

1. to derive the empirical value of c from pure topology;
2. to derive the Planck length independently of c;
3. to identify a fractal index with a directly observed cosmological epoch;
4. to replace the standard observational definition of recombination.

It claims only the following.

Causal Co-Scaling Principle

Distance, time, and the speed of light form a single causal triad:

$$R = ct, \qquad c = \frac{R}{t}, \qquad t = \frac{R}{c}.$$

In a reconstruction hierarchy, one cannot scale distance while leaving time unchanged. Causal admissibility requires co-scaling.

2. The Causal Triad

Let R denote an admissible causal distance, t the corresponding reconstruction time, and c the invariant causal speed. The three elementary relations

$$R = ct, \qquad c = \frac{R}{t}, \qquad t = \frac{R}{c}$$

are formally equivalent, but conceptually they carry different emphases.

If c is viewed as the primitive constant, then R is the distance accessible during time t. If R and t are measured operationally, then c appears as their invariant ratio. In a geometric reconstruction model, neither view alone is sufficient: the causal geometry is the whole triad.

Interpretation. The speed of light is not introduced here as an isolated object. It is the causal ratio binding distance and reconstruction time. The model therefore asks not "Can c be derived from S^3 topology alone?", but rather: "Does the proposed S^3 reconstruction preserve the causal ratio $R/t = c$?"

3. Planck Normalisation and the No-Derivation Caveat

The Planck length and Planck time are

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \qquad t_P = \frac{\ell_P}{c} = \sqrt{\frac{\hbar G}{c^5}}.$$

Thus any model that begins with ℓ_P or t_P has already admitted the causal scale c into its normalisation. Consequently, such a model cannot honestly derive c as an independent result.

Caveat. Using ℓ_P and then obtaining $c = \ell_P/t_P$ is not a derivation of the speed of light. It is a consistency check. The value of c is already present in the Planck normalisation.

The correct target is therefore *causal closure*. A reconstruction hierarchy must preserve the causal relation once the Planck causal unit has been fixed.

4. Axiom of Causal Co-Scaling

We now introduce the central axiom.

Axiom 1 (Causal co-scaling)**.** *Let a reconstruction hierarchy be indexed by $N \in \mathbb{Z}_{\geq 0}$ and let $\varphi = (1+\sqrt{5})/2$. If the carrier radius is scaled from the Planck scale by*

$$R_N = \ell_P \varphi^N,$$

then the corresponding reconstruction time must scale as

$$t_N = t_P \varphi^N.$$

The causal ratio is then invariant:

$$\frac{R_N}{t_N} = c.$$

Proposition 1 (Causal closure)**.** *Under causal co-scaling, every golden-ratio reconstruction level has the same causal ratio:*

$$\frac{R_N}{t_N} = \frac{\ell_P \varphi^N}{t_P \varphi^N} = \frac{\ell_P}{t_P} = c.$$

The proof is algebraic, but the interpretation is structural. The factor φ^N is not permitted to multiply distance alone. If it multiplies the carrier-distance, it must multiply the reconstruction-time. The scaling factor cancels from the ratio, leaving the causal boundary unchanged.

5. Compact S^3 Carrier Volume

If the spatial carrier is a round compact S^3 of radius R, its volume is

$$V_{S^3}(R) = 2\pi^2 R^3.$$

This must be distinguished from the Euclidean ball volume

$$V_{\text{ball}}(R) = \frac{4}{3}\pi R^3.$$

The first is the volume of the three-sphere as a compact three-dimensional manifold. The second is the volume enclosed by an ordinary two-sphere in Euclidean three-space.

For a Planck-normalised golden hierarchy,

$$R_N = \ell_P \varphi^N,$$

the compact S^3 carrier volume becomes

$$V_{S^3}(N) = 2\pi^2 \ell_P^3 \varphi^{3N}.$$

If a fixed carrier content E_0 is distributed across this carrier, then the associated energy-density scale is

$$\rho_E(N) = \frac{E_0}{V_{S^3}(N)} = \frac{E_0}{2\pi^2 \ell_P^3} \varphi^{-3N}.$$

Thus a radius hierarchy scaling as φ^N induces a density hierarchy scaling as φ^{-3N}.

6. Distance, Time, Density, and Reconstruction Capacity

The co-scaling axiom gives a clean separation between causal scaling and density scaling:

$$R_N \sim \varphi^N, \qquad t_N \sim \varphi^N, \qquad \rho_E(N) \sim \varphi^{-3N}.$$

The first two relations preserve the causal ratio. The third expresses geometric dilution of a fixed carrier content across an expanding compact volume. This permits one to introduce a distinct quantity: the number of reconstructible cells or the reconstructible distinguishability capacity. At resolution ϵ, a generic capacity law may be written

$$C_N(\epsilon) \sim \left(\frac{R_N}{\epsilon}\right)^{d_{\text{eff}}},$$

where d_{eff} is an effective reconstruction dimension. At the Planck resolution $\epsilon = \ell_P$,

$$C_N(\ell_P) \sim \left(\frac{R_N}{\ell_P}\right)^{d_{\text{eff}}} = \varphi^{d_{\text{eff}} N}.$$

This gives two opposite monotonic tendencies:

$$\rho_E(N) \sim \varphi^{-3N}, \qquad C_N(\ell_P) \sim \varphi^{d_{\text{eff}} N}.$$

> Energy density and reconstruction capacity are not the same quantity. Energy density measures concentration of carrier content. Reconstruction capacity measures how many distinguishable cells, modes, or states can be supported at a chosen resolution. Expansion may decrease the first while increasing the second.

7. Two Regimes: Pre-Photon Reconstruction and Photon-Supported Expansion

The co-scaling law is most useful when the model distinguishes two regimes.

7.1 Pre-photon reconstruction regime

Before a free-photon description is introduced, one may model the carrier by pure Planck-to-fractal reconstruction:

$$R_N = \ell_P \varphi^N, \qquad t_N = t_P \varphi^N.$$

This is a structural hierarchy. It is not yet an observational expansion law with a measured Hubble parameter. It states how causal distance and reconstruction time must scale if the hierarchy is to preserve the invariant ratio c.

7.2 Photon-supported expansion regime

After the model introduces freely propagating photons, the description must change. A later cosmological regime may involve quantities such as

$$a(t), \qquad H(t), \qquad z, \qquad R(t),$$

and the empirical causal propagation of light. The point of transition between the two descriptions is not defined by the integer alone. It requires a matching condition.

8. Internal Transition Index

Suppose a model chooses an internal index N_* and sets

$$R_* = \ell_P \varphi^{N_*}, \qquad t_* = t_P \varphi^{N_*}.$$

Then by causal co-scaling,

$$\frac{R_*}{t_*} = c.$$

For example, if one takes $N_* = 207$, this gives

$$R_* = \ell_P \varphi^{207}, \qquad t_* = t_P \varphi^{207}.$$

This does not mean that "the integer 207 is recombination." It means only that 207 is an internal reconstruction index, selecting a reference scale inside the model.

Recombination caution. A physical recombination epoch is defined by physical criteria: photon decoupling, temperature, ionisation state, redshift, and optical depth. An internal golden-ratio index may be used as a reference scale, but it does not define recombination by itself. To connect N_* to recombination, one must supply an explicit matching condition.

Numerical caution. Under the pure Planck normalisation of this note, $N_* = 207$ evaluates to

$$R_* = \ell_P \varphi^{207} \approx 2.94 \times 10^8 \text{ m}, \qquad t_* = t_P \varphi^{207} \approx 0.98 \text{ s},$$

i.e. a scale of about one light-second—*not* a recombination scale. A light-travel time of order 3.8×10^5 years would instead require $N \approx 269.6$ under the same simple normalisation. Hence $N_* = 207$ can only be an internal reference index of the reconstruction mechanism; any identification with a cosmological epoch requires an explicit, separately stated matching condition, and none is supplied or claimed here.

A precise statement is therefore:

> If the pre-photon reconstruction mechanism is modeled by golden-ratio scaling from the Planck carrier, then after N_* reconstruction steps the radius reaches the model's chosen pre-photon reference scale. This does not observationally define recombination; it marks a possible point at which the model changes regime from φ-reconstruction to photon-supported expansion.

9. Relation to Previous Causal Self-Consistency Work

A previous draft on causal self-consistency treated the speed of light as an empirical invariant and asked whether a fractal S^3 hierarchy could close consistently under that invariant. The present note extracts the simpler algebraic core of that argument.

The earlier closure form may be summarised schematically as a fixed-point condition: one accepts c as an empirical causal speed and tests whether the proposed hierarchy returns the same value after its internal geometric substitutions. The present note sharpens that idea:

$$R_N = \ell_P \varphi^N, \qquad t_N = t_P \varphi^N \quad \Longrightarrow \quad R_N/t_N = c.$$

Thus the closure is no longer presented as a derivation of c, but as the preservation of the causal triad under co-scaling.

10. Non-Claims

For clarity, the note explicitly rejects five overstatements.

1. **No derivation of c.** Since ℓ_P and t_P contain c, the ratio $\ell_P/t_P = c$ is a consistency identity, not an independent prediction.
2. **No topology-only physics.** Compact S^3 geometry supplies a carrier structure, not a complete derivation of empirical constants.
3. **No direct recombination proof.** A golden-ratio index requires a matching condition before it can be related to photon decoupling.
4. **No replacement of observational cosmology.** The framework is a reconstruction model; observational cosmology remains the test.
5. **No observational support for the φ-hierarchy is claimed.** The directly testable log-periodic signature $\omega = 2\pi/\ln\varphi$ associated with golden-ratio scaling has been tested by us against DESI DR2 BAO and Pantheon+ data with null results,[1] and that status is inherited here: the co-scaling axiom is a structural admissibility condition on a model hierarchy, not an empirically supported scaling law. Nothing in the DRD information-theory line depends on this note.

11. Conclusion

The central result is the axiom of causal co-scaling:

$$R_N = \ell_P \varphi^N, \qquad t_N = t_P \varphi^N, \qquad \frac{R_N}{t_N} = c.$$

It expresses the simple but structurally important fact that distance and reconstruction time cannot be scaled independently once a causal speed is fixed. The golden-ratio factor may organise a reconstruction hierarchy, but it does not alter the causal ratio if time and distance co-scale.

[1] B. Batenin and A. Preece, *A Null Test of Golden-Ratio Log-Periodicity ($\omega = 2\pi/\ln\varphi$) against DESI DR2 BAO and Pantheon+*, research manuscript (2026).

In the broader compact S^3 programme, this axiom supplies a clean interface between three ideas:

$$\text{Planck carrier} \longrightarrow \text{fractal reconstruction} \longrightarrow \text{photon-supported causal expansion.}$$

The principle is modest, but useful. It prevents false derivations of c and replaces them with a sharper demand: every proposed reconstruction scale must preserve the causal triad of distance, time, and light.